**Bimlendra Ray**

• Richardson, Texas (US) • raybimlendra@gmail.com • +16823744919 • LinkedIn • Website

## Education

**University of Texas at Dallas** Richardson, TX
Master Of Science in Mechanical Engineering (DSC). GPA: 3.4 Expected May 2026
Project: Autonomous Vehicles (Current Research Work)
Coursework: Robust Control, Optimal Control, Optimization, Robot Control and Nav.,

Multi-agent system, Modeling and Simulation, MPC, Vehicle Dynamics, Deep learning

**Uttarakhand Technical University** Dehradun, India
Bachelor Of Technology in Mechanical Engineering. 2020
Research work: Review paper on Composite material

**NASA Higher Secondary School** Kathmandu, Nepal
Coursework: Physics. 2016

## Projects

**End-to-End Autonomous Vehicle Control System (Quanser QCar)** UT DALLAS
Head of Autonomous Racing Team Dec 2025 – Mar 2026

- Designed and deployed a real-time autonomous vehicle control architecture integrating perception, planning, and control
- Implemented Model Predictive Control (ACADOS) for real-time trajectory tracking of vehicle lateral and longitudinal dynamics
- Integrated computer vision pipeline (OpenCV / TensorFlow) for traffic light and obstacle detection
- Optimized controller stability and compute latency for real-time execution under hardware constraints
- Developed full closed-loop autonomous navigation system operating in simulation and physical robotic platform.
- Tools: Python, ACADOS, OpenCV, TensorFlow, MPC

**Real-Time Neural Approximation of Constrained MPC via Imitation Learning** UT DALLAS
Feb 2025 – Present

- Designed and implemented a constrained Model Predictive Controller (MPC) for an unstable dynamic system (Linear Inverted Pendulum Model)
- Trained a deep neural network policy using behavior cloning to approximate the MPC feedback law
- Deployed the learned controller in closed-loop simulation and evaluated stability, tracking error, and disturbance rejection
- Analyzed robustness and distribution shift effects in closed-loop learned control
- Tools: PYTHON, TensorFlow, MPC, Optimal Control, Robotics Simulation

**Model Predictive Control for Disturbance-Aware Lap Time Optimization in Autonomous Racing Vehicles using two-step sequential programming** UT DALLAS
Sep 2025 – Dec 2025

- Formulate the lap-time optimization problem under disturbances and model uncertainty.
- Develop an MPC that ensures the actual trajectory stays within a safety region around the nominal optimal racing line.
- Tools: MATLAB, SIMULINK, ACADOS

**Optimal Trajectory following using AL-ILQR algorithms while avoiding dynamic obstacles using chance constraints** UT DALLAS Feb 2025 – May 2025

- Developed an augmented AL-iLQR algorithm enabling optimal trajectory tracking while safely avoiding dynamic obstacles using chance-constraint formulations
- Modeled a linear quadrotor system, discretized the dynamics using RK4, and applied an LTV approximation to ensure real-time feasibility during constrained trajectory tracking.
- Demonstrated reliable obstacle avoidance performance in simulation; identified and analyzed causes of significant deviation from the reference trajectory to guide future controller refinement
- Tools: ILQR, Python

**Detecting and Counting the number of objects on Production Line using PLC** Wilkes University Feb 2024 – May 2024

- Developed PLC program and implemented on real hardware (Allen-Bradley)
- Detecting the number of objects in the container and automatically replacing it with empty using PLC controller
- Setting Alarming system to notify the task completion or any fault occurring during process
- Tools: PLC

**Agri Sort (Sorting Fruits and Vegetable using image classification on moving belt)** UT DALLAS Feb 2025 – May 2025

- Developed a robotic sorting system for fruits and vegetables using ROS and Gazebo.
- Designed and simulated the conveyor belt mechanism.
- Optimized sorting accuracy and reduced manual intervention using AWS Recognition
- Tools: Python, ROS, GAZEBO

## Skills & Interests

- Robotics & Autonomy:
  - MPC, Optimal Control, State Estimation, Robust Control,
  - Autonomous Navigation, Trajectory Planning
- Programming:
  - Python, MATLAB, JavaScript, SQL, PLC
- Robotics & Simulation:
  - SIMULINK, ANSYS, ROS, GAZEBO, QUANSER QCAR
- AI/ML:
  - PyTorch, TensorFlow, Deep Learning, Imitation Learning, Computer Vision
- Design:
  - SolidWorks
- Hardware:

- PLC (Allen-Bradley)

## Experience

**Silicon Soft and IT Consultant Pvt. Ltd.** Kathmandu, Nepal
Software Developer Jan 2022 – Dec 2023

- Gain confidence in learning and doing the task assigned in time
- Improvised the Journey for Mobile Applications development for the first time in this company. Used Kotlin, Jetpack Compose and .Net framework.
- Worked on a Cross-platform Mobile application with other team members. Used SQL, Kotlin, KMM, KTOR
- Developed Government Web application using REACT and JS
- Handled two projects under my responsibilities and led the team member

**Gorkha Brewery Pvt. Ltd. (Carlsberg Group)** Chitwan, Nepal
Maintenance Engineer (Trainee) Jul 2021– Dec 2021

- Implemented Kaizen to find out the root problem in production line and presented a detailed solution
- Performed extended data acquisition and trend analysis to identify system-level energy losses and recommend efficiency improvements

## Leadership & Awards

**GORKHA BREWERY PVT. LTD. (PART OF CARLSBERG GROUP)** Chitwan, Nepal
Kaizen star of the month November 2021 Jul 2021 –Dec 2021

- Modification of the pull of cap Shute to install empty warning alarm sensor to save beer.
- Used the Kaizen tool to find out the root cause of the frequent stoppage of the filler machine and modified the design. This idea increased production by 0.12 million additional bottles per year.